

Assignment

Sr. No. _____

Parameter Estimation

Date: 06/04/26

Q1. let $(x_1, x_2, x_3, \dots, x_n)$ be a random sample of size n taken from a normal pop. with parameters: mean θ_1 and variance θ_2 . Find max likelihood estimates of these 2 parameters.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Ans. $\Rightarrow f(x_i) = \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}}$

$\Rightarrow$ likelihood function:

since obs are independent:

$$L(\theta_1, \theta_2) = \prod_{i=1}^n f(x_i)$$

$$= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}}$$

if separate terms:

$$\Rightarrow \left(\frac{1}{\sqrt{2\pi\theta_2}} \right)^n \cdot e^{-\frac{1}{2\theta_2} \sum (x_i - \theta_1)^2}$$

$$= (2\pi\theta_2)^{-n/2} e^{-\frac{1}{2\theta_2} \sum (x_i - \theta_1)^2}$$

Taking log likelihood

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$$\ln L = \ln[(2\pi\sigma_2^2)^{-n/2}] + \ln \left[e^{-\frac{1}{2\sigma_2^2} \sum (x_i - \mu_1)^2} \right]$$

$$= -\frac{n}{2} \ln(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} \sum (x_i - \mu_1)^2$$

$$= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma_2^2) - \frac{1}{2\sigma_2^2} \sum (x_i - \mu_1)^2$$

differentiating w.r.t. μ_1 .

$$\frac{\partial \ln L}{\partial \mu_1} = -\frac{1}{2\sigma_2^2} \cdot \frac{\partial}{\partial \mu_1} \sum (x_i - \mu_1)^2$$

$$\text{now :- } \frac{\partial (x_i - \mu_1)^2}{\partial \mu_1} = -2(x_i - \mu_1)$$

$$\begin{aligned} \text{so } \frac{\partial \ln L}{\partial \mu_1} &= -\frac{1}{2\sigma_2^2} \cdot (-2) \sum (x_i - \mu_1) \\ &= \frac{1}{\sigma_2^2} \sum (x_i - \mu_1) \end{aligned}$$

setting it to 0 ;

$$\sum (x_i - \mu_1) = 0$$

$$\sum x_i - n\mu_1 = 0$$

$$\mu_1 = \frac{1}{n} \sum x_i = \bar{x}$$

now differentiating with r.t. θ_2 .

$$\frac{\partial \ln L}{\partial \theta_2} = \frac{-n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum (x_i - \theta_1)^2.$$

setting $\Rightarrow 0$. $\frac{-n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum (x_i - \theta_1)^2 = 0$

multiplying LHS ; RHS by $2\theta_2^2$.

$$-n\theta_2 + \sum (x_i - \theta_1)^2 = 0.$$

$$\sum (x_i - \theta_1)^2 = n\theta_2.$$

$$\theta_2 = \frac{1}{n} \sum (x_i - \theta_1)^2.$$

substitute $\theta_1 = \bar{x}$.

$$\theta_2 = \frac{1}{n} \sum (x_i - \bar{x})^2.$$

$$\hat{\theta}_1 = \bar{x}, \quad \hat{\theta}_2 = \frac{1}{n} \sum (x_i - \bar{x})^2.$$

- Q. let $x_1, x_2, \dots, x_n$ be a random sample from $B(n, \theta)$ distribution where $\theta \in (0, 1)$ is unknown and 'n' is a known +ve int. compute value of θ using Maximum Likelihood Estimation

Ans. assuming an estimator of θ as

$$\hat{\theta} = a\bar{x} \quad \text{where } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\Rightarrow \text{Mean square Error} = E[(\theta - \hat{\theta})^2] \\ = E[(\theta - a\bar{x})^2]$$

expanding expression -

$$\Rightarrow E[\theta^2 - 2a\theta\bar{x} + a^2\bar{x}^2]$$

$$\Rightarrow \theta^2 - 2a\theta E[\bar{x}] + a^2 E[\bar{x}^2]$$

now for binomial :- (i) mean of x_i :

$$E[x_i] = n\theta$$

$$\text{so } E[\bar{x}] = \frac{1}{n} \sum E[x_i] = \frac{1}{n} \cdot n(n\theta) = n\theta$$

(ii) $\Rightarrow$ Variance of x_i :-

$$\text{Var}(x_i) = n\theta(1-\theta)$$

$$\text{so } \text{Var}(\bar{x}) = \frac{\text{Var}(x_i)}{n} = \frac{n\theta(1-\theta)}{n}$$

(iii) Find $E[\bar{x}^2]$:-

$$E[\bar{x}^2] = \text{Var}(\bar{x}) + (E[\bar{x}])^2 \\ = \frac{n\theta(1-\theta)}{n} + (n\theta)^2$$

substituting in MSE :

$$= \theta^2 - 2a\theta(n\theta) + a^2 \left[\frac{n\theta(1-\theta)}{n} + (n\theta)^2 \right] \\ = \theta^2 - 2an\theta^2 + a^2 \frac{n\theta(1-\theta)}{n} + a^2 n^2 \theta^2$$

differentiating w.r.t a .

$$\frac{d(MSE)}{da} = -2m\theta^2 + \frac{2am\theta(1-\theta)}{n} + 2am^2\theta^2.$$

setting to 0.

$$-2m\theta^2 + \frac{2am\theta(1-\theta)}{n} + 2am^2\theta^2 = 0.$$

$$\Rightarrow -m\theta^2 + \frac{am\theta(1-\theta)}{n} + am^2\theta^2 = 0.$$

$$\Rightarrow a \left(\frac{m\theta(1-\theta)}{n} + m^2\theta^2 \right) = m\theta^2.$$

$$a = \frac{m\theta^2}{\frac{m\theta(1-\theta)}{n} + m^2\theta^2}.$$

~~for $\theta = 0$~~ $\Rightarrow a = \frac{0}{\frac{1-\theta}{n} + m\theta}$

now to get a usable parameter.

$$\hat{\theta} = \frac{\bar{X}}{m}, \text{ as } E\left[\frac{\bar{X}}{m}\right] = \theta.$$

$$\text{so } \hat{\theta}_{MMSE} = \frac{\bar{X}}{m}.$$

$$= \frac{1}{mn} \sum x_i.$$